

Electric Conduction

1. By how much does the emitted current change if the tungsten cathode temperature is increased from $T = 2200\text{ K}$ to 2350 K ? The work function for tungsten is $W_{\text{ki}} = 4.5\text{ eV}$. The Richardson coefficient is $6.02 \cdot 10^5\text{ A/m}^2\text{K}^2$.
2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at $T = 2200\text{ K}$? The electron charge is $q = -1.6 \cdot 10^{-19}\text{ As}$.
3. What is the free-electron drift velocity in a copper conductor of radius $R = 1.5\text{ mm}$ if the current is $I = 12\text{ A}$? The electron density in copper is $n_e = 3.3 \cdot 10^{28}\text{ m}^{-3}$.
4. We examine an n-type semiconductor under the conditions shown in the figure: $B_y = 0.4\text{ T}$, $U_{\text{H}} = 10\text{ mV}$, $I = 30\text{ mA}$. The magnetic induction is directed along the y -axis.
 - (a) What is the electron concentration?
 - (b) What is the drift velocity of the electrons?
 - (c) What is the Hall constant?